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Design and construction of a smart battery charger

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Abstract

This article presents the design and construction of a smart battery charger that ensures efficient, safe, and reliable charging of a 12 V lead-acid battery. The aim is to design and implement a smart charger with adaptive charging algorithms, on-site/real-time monitoring, and IoT-based remote monitoring. The system used an ESP32 microcontroller to detect battery voltage, adjust charging current, and disconnect the charger once the battery is fully charged. It integrated voltage, current and temperature sensors to prevent overcharging, overheating, and other faults that reduce battery lifespan. An IoT feature enabled remote monitoring and status updates through the Blynk app using Wi-Fi connectivity. The methodology involves an ESP32 microcontroller, buck converter, voltage and current sensors, relay module, LCD, and buzzer. The system operates in constant voltage mode only with automatic cut-off at 14.4V. Testing showed the battery voltage rising from 11.8 V to 14.4 V with a gradual current decrease, confirming correct charging behaviour. Overall, the system provides a practical, low-cost, and intelligent smart battery charger suitable for unstable power conditions.

Keywords Battery, Buck-Converter, Current Sensors, Microcontroller, IoT, Smart, Voltage, Sensors

1 Introduction

A smart battery charger is defined as an advanced charging system meant to automatically check and regulate the charging and discharging processes of batteries. Unlike traditional chargers, it utilizes microcontrollers configurations to regulate voltage and current depending on the battery's state, avoiding overcharging, overheating, and prolonging battery life span. The design and construction of such a charger characteristically involve integrating sensors, control algorithms, and shielding features to guarantee safe, efficient, and unswerving energy management. Steady electricity supply is the backbone of developed country, but when the reverse becomes the everyday experience, economy of the country crawls. Globally, electrical power supply is not equal to power demand due to several factors such as increase in populations, corruptions, insecurity, vandalism, political interest and so on. Inconsistent power supply is almost in every country in



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Africa as well as other part of the globe but Nigeria's own is the champion. And because of that, many people find different options in storing electrical energy in case of power outage. In many parts of Nigeria, electricity supply is often unpredictable, with frequent outages and voltage fluctuations disrupting daily life. As a result, households and small businesses rely heavily on rechargeable 12 V lead-acid batteries to keep essential devices running. However, using conventional chargers often leads to problems such as battery swelling, overheating, or loss of capacity due to overcharging or undercharging [1]. Most traditional chargers operate at fixed rates and provide no feedback on the battery's condition, leaving users unaware of potential risks. This situation highlights a clear need for smarter, safer, and more adaptive battery charging solutions [2].

For example [3], proposed a microcontroller-based smart battery charger capable of monitoring real-time voltage and current and displaying this information to the user. Their study showed how microcontrollers can enhance user interaction and control, but the design lacked automatic cut-off and sufficient protective features, making it vulnerable to overcharging during unstable power conditions. Similarly [4], highlighted that intelligent chargers using adaptive algorithms could improve battery lifespan by adjusting current and voltage based on the battery's condition. These studies suggest that conventional fixed-rate chargers are inefficient in the long term because they fail to consider critical factors such as battery temperature, charge depth, and real-time response. Recent developments in smart charging have further expanded the possibilities [5]. developed a solar-powered charge controller using Maximum Power Point Tracking (MPPT) to optimize energy extraction [6]. introduced a hybrid charger that switches between solar and grid power depending on availability, improving reliability in regions with inconsistent sunlight. Although effective, these designs are often complex and expensive, limiting adoption in low-income areas.

The introduction of IoT in battery charging has added a new level of convenience and control. Studies like [7] and [8] demonstrate chargers that allow users to remotely monitor battery voltage, current, and charging status via smartphones or cloud platforms. While these systems increase user awareness, they rely on stable internet access and consume additional power, which may be challenging in regions with unreliable electricity.

Safety is also an important consideration [9]. designed a temperature-controlled lithium-ion charger that halts charging if overheating occurs, reducing risk. Advanced adaptive algorithms, such as fuzzy logic and predictive intelligence, have been explored by to improve charging efficiency and prevent hazards like dendrite formation. However, these approaches increase complexity and cost, making them less practical for simple consumer applications in developing countries [10].

From this review, a clear gap emerges: few chargers combine real-time sensing, adaptive CV charging, automatic cut-off, user feedback, and IoT monitoring into a single, affordable, and practical device. There is a need for a smart, low-cost charger that provides safe and reliable charging for 12 V lead-acid batteries, which are widely used in Nigerian homes and small businesses.

This research aims to fill that gap by designing and building a microcontroller-based smart battery charger. The system will include voltage and current sensing, CV charging control, an automatic relay cut-off, an LCD display, a buzzer alert, and IoT monitoring through the Blynk platform. By integrating these features into a simple and cost-effective

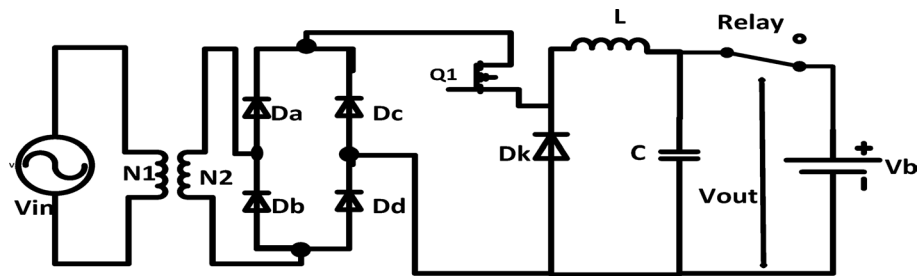


Fig. 1 Circuit diagram of the system

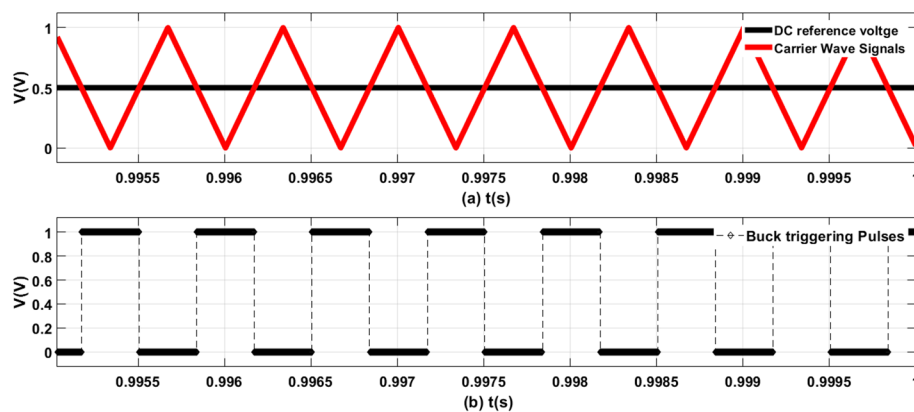


Fig. 2 a: Comparison of DC reference signal and Carrier wave. b: Buck converter Triggering Pulses

design, the charger will offer safer, more intelligent, and reliable battery charging, helping households and small businesses cope with frequent power interruptions.

2 Methodology

This project uses experimental and theoretical investigation, where the smart battery charger is built, programmed, and tested in real conditions to see how it performs. The practical work is supported with basic theoretical calculations to guide the design and ensure the circuit works correctly. The circuit diagram of the system is shown in Fig. 1 with input voltage, V_{in} (220Vrms), transformer ratio of $N1$ and $N2$, power diodes (D_a , D_b , D_c and D_d). It is made of AC power source, stepdown transformer, rectifier, DC-DC buck converter ((Mosfet $Q1$, inductor L , capacitor, C and Power diode, D_k), relay, and battery bank. The smartness of the proposed system is based on the sensing the voltage, current and temperature as well as in their regulations using microcontroller-based system. The power source supplies current to the rectifier. The rectifier, converts AC current to DC current, and delivers the current to the buck converter. The buck converter steps down voltage and raising the current. When relay connects to the battery. The voltage, current and temperature sensors communicate to the microcontroller on the state of battery based on the level of voltage, current condition and the prevailing temperature.

In this project, the buck converter controlled using a simple pulse width modulation (PWM) technique. The PWM signal is generated by comparing a reference signal representing the required output level with a high-frequency carrier waveform as represented in Fig. 2a. It is observed that the carrier wave signal has amplitude of 1.0 V. and

frequency of 15 kHz; and the DC voltage signal is 0.5 V. Whenever the reference signal is higher than the carrier, the output pulse goes HIGH (1), and when it is lower, the pulse goes LOW (0) as shown in Fig. 2b. By switching repeatedly between 0 and 1, the system adjusts the duty cycle of the switch and controls how much power is delivered to the battery during charging [11].

The waveform shown in Fig. 2b represents the PWM triggering pulses that control the switching operation of the buck converter in the smart battery charger. These pulses alternate between HIGH (1) and LOW (0) at a fixed switching frequency. Each HIGH–LOW transition determines when the MOSFET switch, Q inside the buck converter turns ON or OFF. This rapid switching is what allows the converter to step down the input voltage to a safe charging voltage for the battery.

When the pulse is HIGH, the MOSFET turns ON and current flows from the supply into the inductor. During this ON period, the inductor stores energy and the output voltage starts to rise. When the pulse goes LOW, the MOSFET, Q turns OFF, and the inductor, L releases its stored energy to the battery through the diode. This OFF period helps maintain continuous current and prevents the output from dropping abruptly.

By adjusting the duty cycle that is, the percentage of time the pulse stays HIGH in each switching cycle the buck converter can precisely regulate the battery's charging voltage and current. A higher duty cycle delivers more energy to the battery (useful when the battery is low), while a lower duty cycle reduces the energy delivered as the battery approaches full charge [12].

Overall, these triggering pulses act as the “heartbeat” of the buck converter, ensuring smooth, controlled, and efficient charging. They allow the smart battery charger to automatically shift between constant-current (CC) and constant-voltage (CV) modes while maintaining stability and protecting the battery from overcharging [13].

3 Design calculation of the proposed system

3.1 Mathematical analysis of the AC-to-DC power converter

The peak secondary voltage of the supply power of the proposed topology is expressed as in Eq. (1) [14]

$$V_{pk} = V_{rms} \times \sqrt{2} = V_{in} \times \sqrt{2} \quad (1)$$

Where V_{pk} and V_{rms} are the peak secondary voltage and the root-mean-square voltage respectively.

The output voltage of the bridge rectifier is written as in Eq. 2 [15]:

$$V_{DC} = V_{peak} - 2V_D \quad (2)$$

V_{DC} and V_D are the rectifier output voltage and the voltage across the power diode.

The relationship between the secondary current, root-mean square current and input power is written in Eq. (3) [15]

$$I_{sec}(rms) = \frac{P_{in}}{V_{rms}} \quad (3)$$

$I_{sec}(rms)$ – secondary current, P_{in} – input power

The output frequency of the single-phase H-bridge rectifier is twice the source frequency and it is expressed in Eq. 4 [16].

$$\text{Ripple Frequency : } f_{out} = 2 \times f_s \quad (4)$$

$$\text{Filter Capacitance : } C = \frac{\Delta I_{load}}{f_{out} \times \Delta V} \quad (5)$$

f_{out} , f_s , C , ΔI_{load} , ΔV are output frequency, source frequency, change in current and change in voltage.

3.2 Buck converter operation

When the AC voltage is rectified and filtered, the high voltage is brought down by the switching action of the power Mosfet, and the regulation of the charging action is done by the adjustment of duty cycle, D, period, T, turn on time, t_{on} , and the change in inductor current. The expression of duty cycle is presented as in Eq. 6.

a) Duty Cycle of PWM:

$$D = \frac{t_{on}}{T} \quad (6)$$

t_{on} and T are turn on time and period respectively

The output voltage of buck converter and inductor ripple current, Δi_L are expressed as in Eqs. (7) and (8).

Output Voltage of Buck Converter:

$$V_{out} = D \cdot V_{in} \quad (7)$$

$$\Delta i_L = \frac{[(V_{in} - V_{out}) \times D]}{L \times f_s} \quad (8)$$

The Eq. 7 shows that the output voltage depends on the duty cycle and the input voltage (V_{in})

Also Eq. 8 tells us how much the current through the inductor fluctuates (ΔI_L), which is important for smooth charging.

Moreover, to determine the actual inductance and the capacitance of the DC-DC buck converter, the Inductor Peak and Minimum Currents are first presented in Eqs. (9) and (10), considering that the average inductor current is equal to the current flowing to the load ($I_{L(average)} = I_{out}$) [17]

$$I_{L(peak)} = I_{out} + \frac{\Delta i_L}{2} = \frac{V_o}{R} + \frac{V_o [1 - D] T}{2L_{peak}} \quad (9)$$

$$I_{L(min)} = I_{out} - \frac{\Delta i_L}{2} = \frac{V_o}{R} + \frac{V_o [1 - D] T}{2L_{min}} \quad (10)$$

Under lossless and continuous current mode (CCM) conditions, $I_{L(min)} = 0$, the minimum inductance is written as in Eq. 11

$$L_{min} = \frac{[1 - D] R}{2f_s} \quad (11)$$

R- load resistance.

Then, the actual inductance of the inductor of the DC-DC buck converter becomes

$$L = L_{min} + 25\%L_{min} = 1.25L_{min} = 1.25 \frac{[1 - D] R}{2f_s} \quad (12)$$

The $25\%L_{min}$ is added to the minimum inductance to ensure that the system does not enter into discontinuous current mode (DCM).

In a similar way, actual capacitance of the capacitor of the DC-DC buck converter is expressed as

$$C = C_{min} + 25\%C_{min} = 1.25C_{min} = 1.25 \frac{[1 - D] V_o}{8f_s^2 L}$$

The output voltage ripple depends on the current ripple (Δi_L), switching frequency (f_s), and capacitance of capacitor C.

Further, for PWM Triggering Condition, the logic expression of it is presented in Eq. 13

$$Trigger\ pulse = \begin{cases} 1, & V_{ref} > V_c \\ 0, & V_{ref} < V_c \end{cases} \quad (13)$$

V_{ref} and V_c are the DC reference voltage and carrier signal voltage.

A block diagram presented in Fig. 3 illustrates the complete system operation of the IoT Smart Batter.

charger, showing how each unit of the system is interconnected to achieve safe, efficient, and intelligent battery charging. This diagram serves as the central reference for understanding how power flows from the AC mains, how the charging at constant voltage is regulated, and how the microcontroller monitors system parameters during operation. By examining the layout, the relationship between the power stage, control stage, sensing units, protection components, and IoT communication features becomes easy to trace.

The block diagram begins with the 220 V AC mains, which is stepped down by the transformer to a safer 15 V AC. This is rectified by a full-bridge diode arrangement and then filtered using a capacitor to produce a smoother DC voltage. From this point, the LM2596 buck converter regulates the output to the required 14.4 V charging voltage, ensuring stable operation during Constant Voltage (CV) charging mode. A separate regulated 5 V output supplies the ESP32 microcontroller, sensors, LCD display, relay module, and buzzer, ensuring that all low-voltage components operate safely and reliably. The voltage divider network and ACS712 current sensor feed real-time voltage and current values to the ESP32. These readings enable the microcontroller to intelligently control the relay, automatically disconnecting the battery once the full-charge voltage is reached. The ESP32 also sends the measured parameters to the Blynk IoT platform, allowing the user to remotely monitor battery status from a smartphone.

Overall, the block diagram provides a clear and complete representation of how the hardware components interact to form a functional smart charger. It highlights the flow of signals, the power regulation stages, the safety features, and the integration of IoT technology. This makes Fig. 3 an essential visual guide for understanding the operation and implementation of the system.

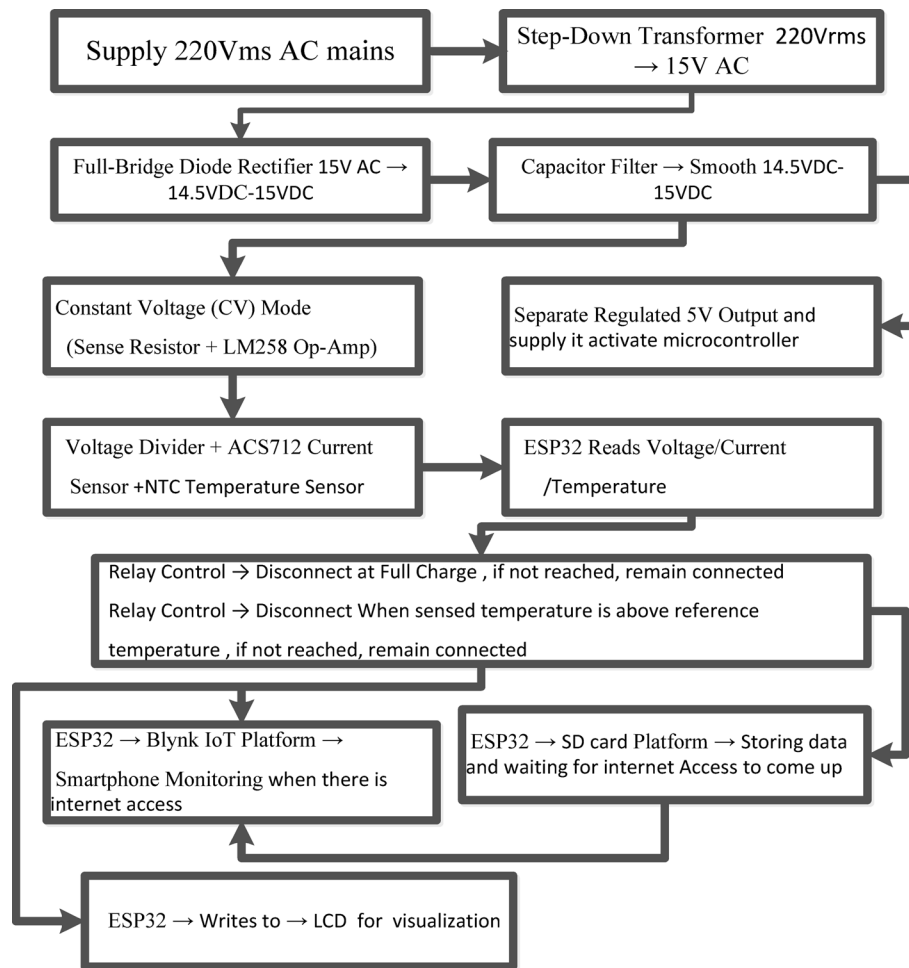


Fig. 3 Block diagram of Smart Battery Charger

Under internet loss, the state of the system operation is stored in SD card pending on the return of internet access. After internet restoration, the IOT is updated from the information stored in SD card.

4 Result and discussion

The operation of the smart battery charger was further examined using the simulation waveforms generated during testing. These waveforms include the PWM signal, inductor current, and output voltage, and they collectively demonstrate how the converter transitions between the constant-current and constant-voltage charging stages. By studying these graphs, the behaviour and stability of the circuit can be clearly understood.

The current waveform shown in the Fig. 4a represents the inductor current of the buck converter. Its shape resembles a fluctuating triangular waveform, and this form is completely normal for converters that operate in Continuous Conduction Mode (CCM). In CCM, the current flowing through the inductor never drops to zero at any point during the switching cycle. Instead, it continuously rises and falls as the switch turns ON and OFF. When the PWM signal turns the switch ON as shown in Fig. 4(c), the input voltage is applied across the inductor at $0 \leq t \leq 0.01$ at range of 40 V–25 V. This causes the inductor current to rise linearly while charging and falls linearly, while discharging, because

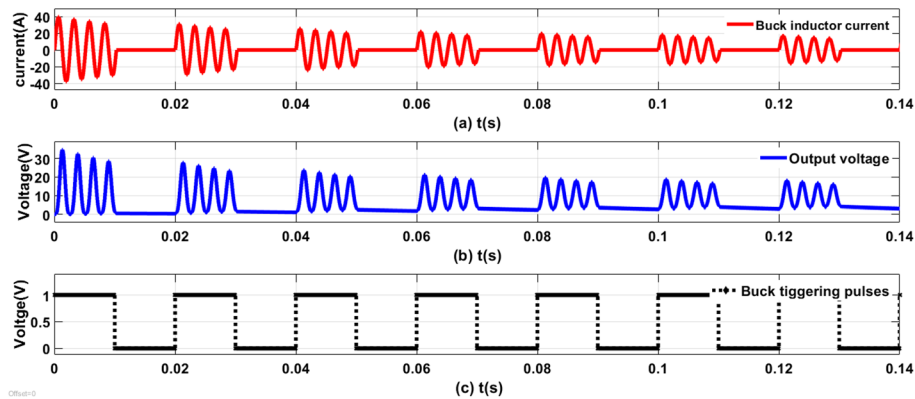


Fig. 4 (a) Buck inductor current. (b) Output charging voltage. (c) Buck Triggering pulses

the inductor is storing magnetic energy during charging and releasing magnetic energy during discharging. When the PWM signal turns the switch OFF, the inductor releases this stored energy entirely to the load (the battery), causing the current to fall linearly but not to zero. This repeated rising and falling forms the triangular pattern seen in the waveform. The small oscillations or “ringing” at the switching edges occur because of the interaction between the inductor and small parasitic capacitances inside the circuit. These oscillations are normal and quickly settle down, showing that the converter is stable and not experiencing harmful resonance.

This shape of the current waveform is important because it confirms that the buck converter is operating efficiently and supplying a smooth current to the battery. Charging batteries requires controlled current flow to prevent damage, and the triangular current waveform indicates that the converter is regulating the charging current properly through constant-voltage (CV) charging phase. The output voltage is seen in Fig. 4b and follows the pattern of triggering pulses. Figure also shows the charging voltage of 15 V.

The output voltage waveform shown in the simulation represents the DC charging voltage delivered by the buck converter to the battery. This waveform is smoother than the current waveform because the converter uses an output capacitor to filter out most of the switching fluctuations. As a result, the voltage waveform appears mostly steady with only small ripples.

The slight rising and falling of the voltage during each switching cycle occurs because the converter alternates between the MOSFET being ON and OFF. When the PWM signal.

turns ON, the inductor receives energy and the output voltage slightly increases. When the PWM signal turns OFF, the inductor releases energy through the diode, causing the voltage to slightly decrease.

These small variations form the low-ripple voltage waveform that is typical of a buck converter. The capacitor smooths these ripples by storing charge during the ON period and releasing it during the OFF period, which keeps the voltage stable.

The reason this waveform is important is that batteries require a steady and controlled voltage during the Constant Voltage (CV) phase of charging. The smoothness of the waveform shows that the converter is functioning correctly and providing a clean DC output suitable for safe charging. The presence of only minor ripple means the battery receives a stable voltage, which helps prevent overheating, overcharging, and damage to the battery plates.

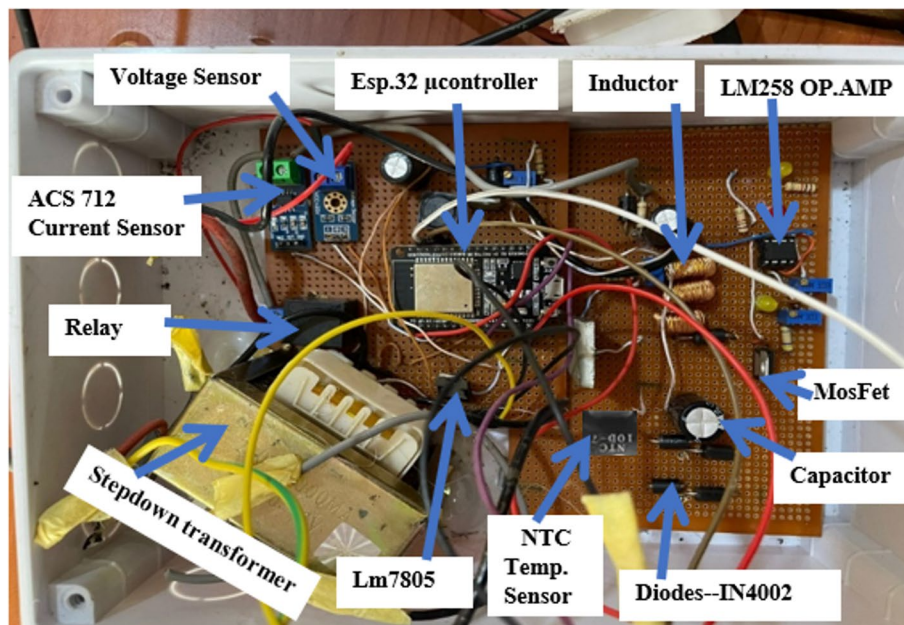


Fig. 5 A Complete Hardware Setup of the Smart Battery Charger



Fig. 6 Picture of LCD Display during Operation

Figure 5 displayed the experimented step-up of the test system. It consists of real time AC supply system, the step-down transformer, the single-phase uncontrolled AC-DC power converter, current sensor module, voltage sensor module, ESP 32 Microcontroller, NTC temperature sensor, communication module and other logic circuits a. The test system converts AC supply to DC power. The DC power is further processed to feed the logic circuits and supply to DC-DC power converter that charges the battery. The performance of the charging system and the condition of the battery is communicated the Blynk App that informs the user the operation of the system.

LCD screen attached to the experimented system is shown in Fig. 6. It is observed that the current and voltage were 7.7 A and 13.1 V on plastic white box.

Figure 7 shows the results from the Blynk app. It is observed that he battery voltage and current are 13.01 V and 8.22 A respectively.

In furtherance, it is nice to emphasize that the complete hardware setup in Fig. 5 integrated the microcontroller unit, the power conversion stage of the smart charger, the

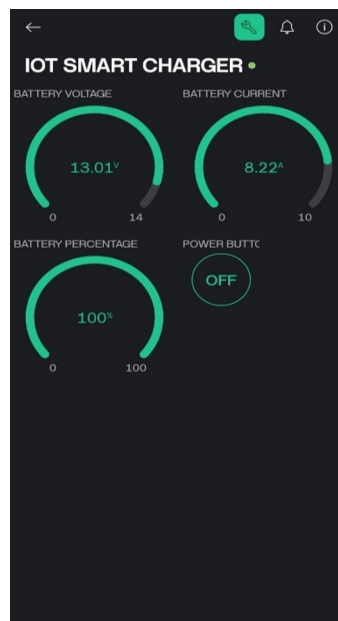


Fig. 7 Screenshot of Blynk App Display Showing Voltage, Current, and Status

sensing elements, and the user-interface components into a single functional system. All modules are interconnected to allow seamless measurement, processing, and display of battery charging parameters during operation.

During live operation, the LCD display in Fig. 5 serves as the primary local monitoring interface. It continuously presents real-time values such as the battery voltage, charging current, the battery and so on charging status. This enables the user to visually observe the behaviour of the charger directly on the device, ensuring immediate feedback on the battery's charging progress.

Simultaneously, the ESP-based controller transmits the same measured parameters to the Blynk IoT platform via Wi-Fi. The Blynk application displays these values on a structured dashboard, updating in real time as the charger transitions between different operational states as seen in Fig. 6. Through this interface, the user can remotely monitor key indicators such as voltage, current, temperature, battery percentage, and charging status from any smartphone connected to the internet.

This combination of local display through the LCD module and remote monitoring through the Blynk app enhances the usability of the system. It provides both on-device visibility and IoT-enabled supervision, making the hardware setup more interactive, convenient, and suitable for a smart battery charging environment.

Figure 8 presents a comparative analysis between the triangular carrier waveform and the direct current (DC) reference signal. The carrier waveform is characterized by an amplitude of 12.5 V and a frequency of 14.7 kHz, whereas the DC reference signal maintains a constant amplitude of 12.5 V. The superposition and comparison of these two signals result in the generation of a pulse-train waveform, as depicted in Fig. 9.

Figure 9 illustrates the laboratory-generated buck converter triggering pulses employed for the switching operation of the power MOSFET, designated as Q1. These pulses exhibit a frequency of 14.15 kHz and attain a maximum amplitude of 14.10 V. In the absence of these triggering signals, the power MOSFET remains in the OFF state and cannot be activated.

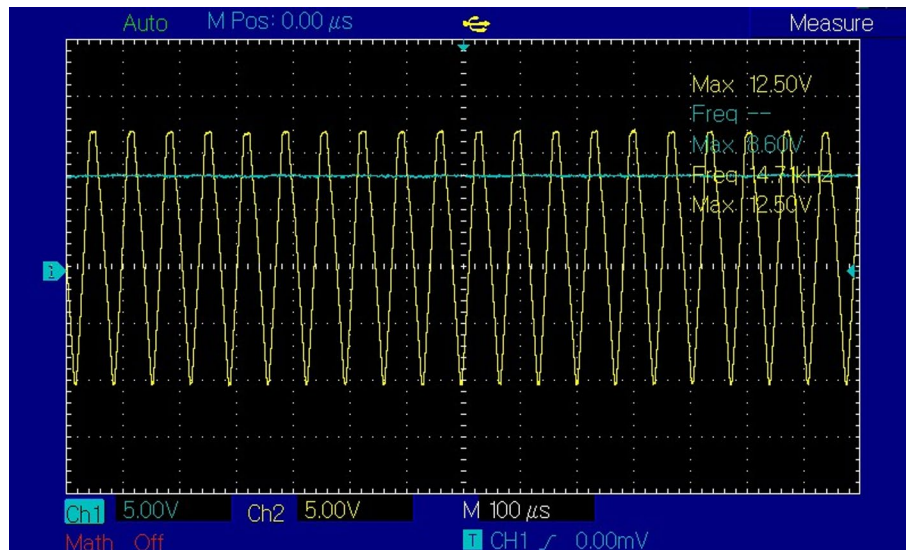


Fig. 8 Comparison of the triangular wave and DC wave reference signal

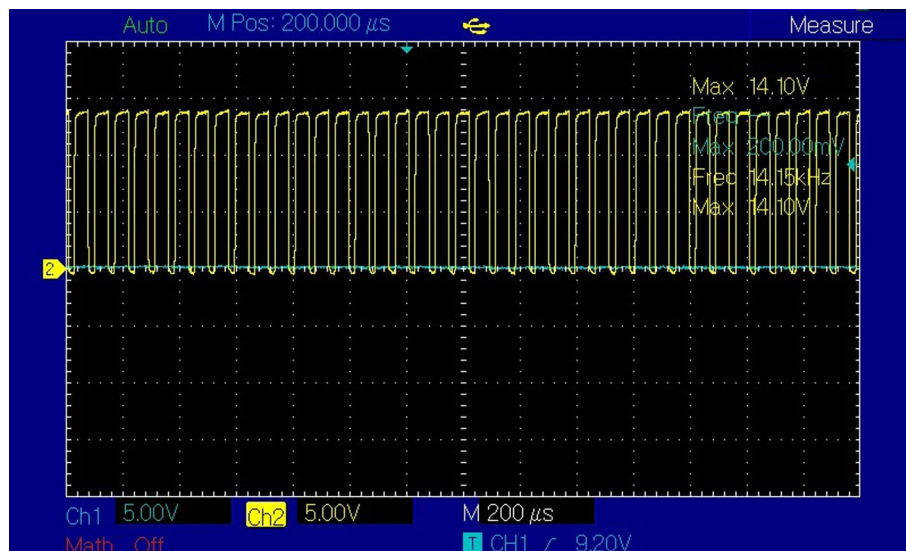


Fig. 9 Laboratory buck triggering pulses

Figure 10 illustrates the single-phase alternating current (AC) supply, which is characterized by a frequency of 50 Hz and a voltage magnitude of 328.68 V. This input serves as the primary source for the step-down transformer. The transformer plays a critical role in reducing the supply voltage to a lower, more manageable level suitable for subsequent conversion and utilization within the system. The step-down process not only ensures compatibility with downstream circuitry but also enhances operational safety.

Figure 11 demonstrates the output of the step-down transformer, which reduces the input supply to a voltage magnitude of 36.83 V while maintaining the operating frequency at 50 Hz. This transformation is essential for adapting the higher input voltage to a level suitable for subsequent conversion and utilization within the system. The reduction to 36.83 V not only ensures compatibility with downstream components but also contributes to improved safety.

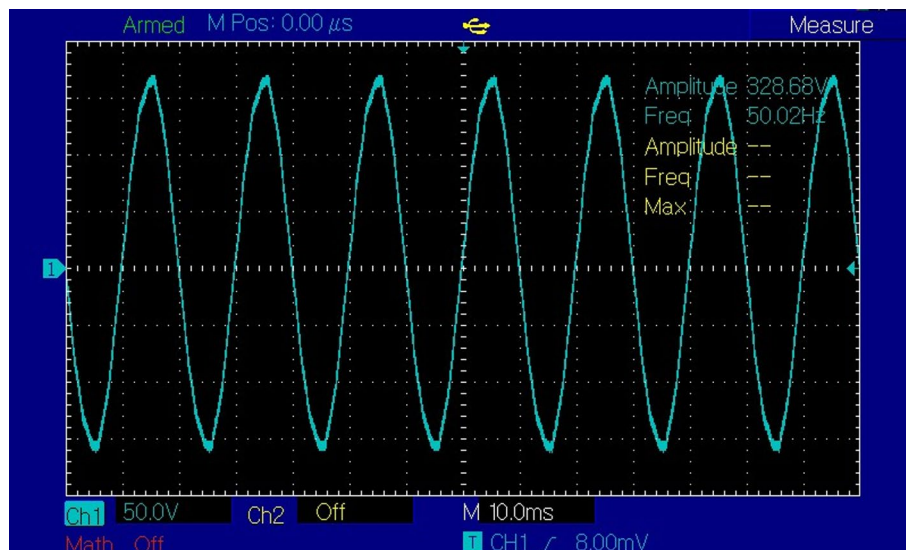


Fig. 10 Single-phase AC voltage supply

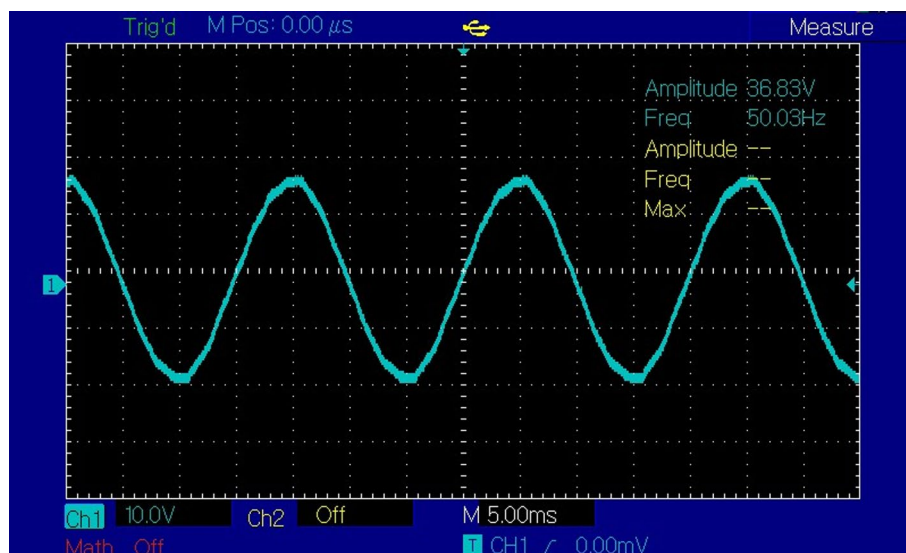


Fig. 11 Stepdown AC voltage output of transformer supply

The output voltage of the transformer is subsequently supplied to a single-phase AC–DC uncontrolled rectifier, which generates the rectified waveform depicted in Fig. 12. As observed in the figure, the rectifier produces an output with a voltage magnitude of 36.43 V and a rectified frequency of 100.46 Hz. This frequency doubling effect is characteristic of full-wave rectification, where the negative half-cycles of the input waveform are inverted to contribute to the output. The resulting waveform demonstrates the effective conversion of the AC supply into a pulsating DC signal, thereby providing the necessary foundation for further filtering and regulation stages within the system.

Figure 13 depicts the voltage waveform that is applied to the input of the DC–DC buck converter. The converter effectively reduces the input voltage to a range between 13.80 V and 15.0 V, which is suitable for charging the battery. This voltage regulation is critical, as it ensures that the charging process occurs within the safe operating limits of the

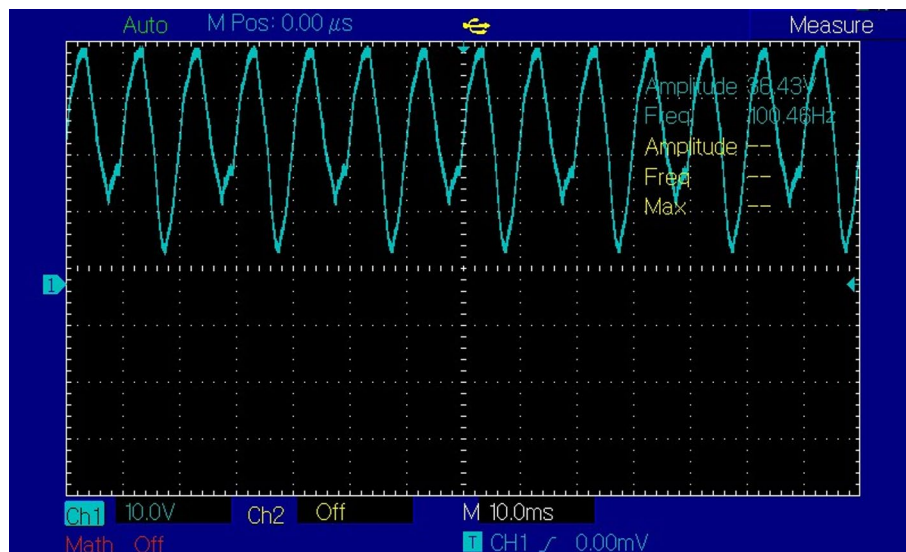


Fig. 12 Rectified AC voltage of the test system

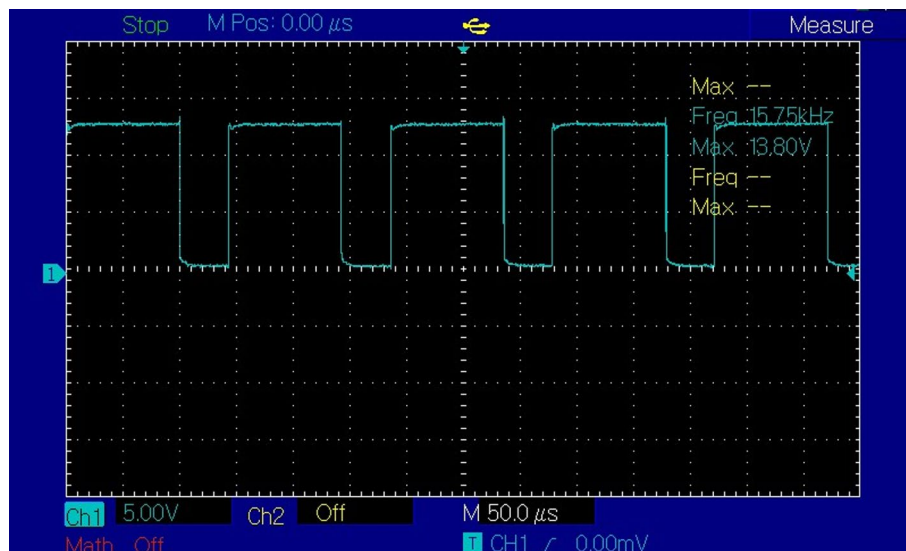


Fig. 13 Charging Voltage of the test system

battery, thereby preventing overvoltage stress and enhancing both the reliability and longevity of the energy storage system. The observed performance of the buck converter demonstrates its capability to provide a stable and controlled output, highlighting its importance in efficient power management within the overall system.

Figure 14 presents the oscilloscopic representation of the battery voltage during the charging process. The waveform indicates a stable voltage level of 13.86 V, confirming that the DC–DC buck converter successfully regulates the output within the desired charging range. This observation demonstrates the converter’s effectiveness in maintaining a consistent voltage supply, which is essential for ensuring safe battery operation, preventing overcharging, and prolonging the overall lifespan of the energy storage system.

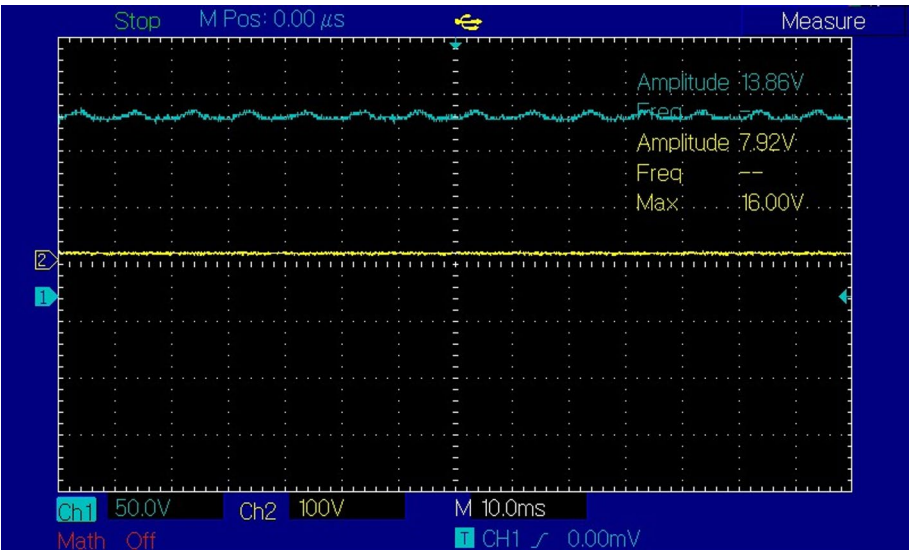


Fig. 14 Battery voltage

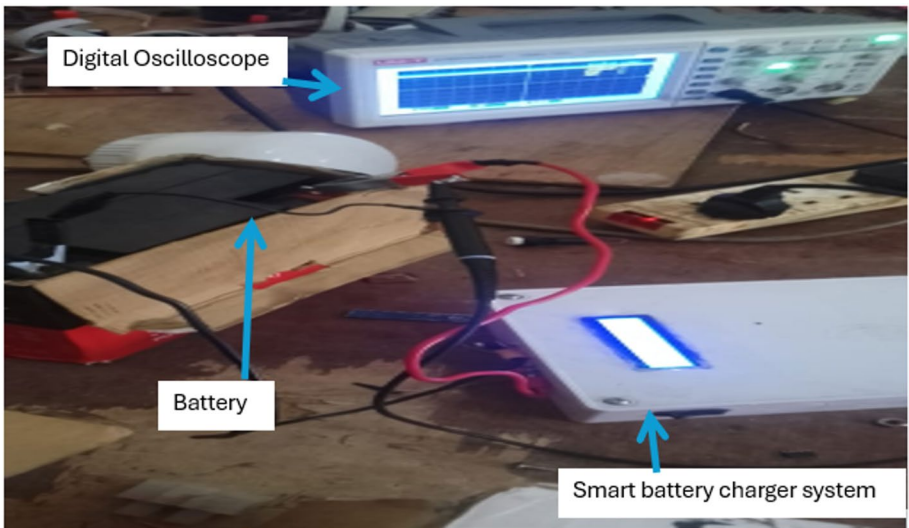


Fig. 15 Overall practical set-up of the experimented system

Figure 15 presents the overall practical set-up of the experimental system. The configuration comprises the battery under test, a digital oscilloscope for waveform monitoring, and the smart battery charging system. This arrangement provides a comprehensive view of the experimental environment, enabling both the observation of voltage and current characteristics and the evaluation of the charging performance. The inclusion of the oscilloscope ensures accurate measurement and validation of the system’s operation, while the smart charging unit demonstrates the integration of control and regulation mechanisms essential for efficient energy management.

The parameters of the research system are summarized in Table 1. These parameters provide a comprehensive overview of the operating conditions and design specifications that govern the performance of the system.

The comparison between the simulated results and the experiment results of research system are shown in Table 2.

Table 1 Research system Parameters

Items	Ratings/results
Capacitances, C	0.36 μF
AC supply Voltage	220-230VAC
Buck Voltage	14.40–14.5 V
Battery Capacity	18AH, 12 V
Experimental Duty ratio D	0.960
Power MosFet	IORT P4360
Current Sensor and voltage sensor	ACS 712 Current and Dc VCC < < 25 V
Regulators	LM 7812 V and LM 705 V
Transformer	220VAC, 1000 mA, 15-0-15 V
Operational Amplifier	LM 258 OPAMP
NTC Temperature Sensor	NTC 10D-7
Capacitor after rectifier	220 μ F, 50 V
Inductor and load resistance	0.05mH and 10 Ω
Operating frequency	49.99–50 Hz
Switching frequency	14–15 kHz
Buck capacitance and ripple	0.0024 μF and 0.01
Rectifying Diodes	IN4002

Table 2 comparison of some simulated results and the experiment results

Items	Simulated Results	Experimented Results
Supply voltage	220Vrms (311.12 V)	328.68 V
Carrier frequency	15 kHz	14.7 kHz
Supply frequency	50 Hz	50.02 Hz
Duty ratio of buck converter	0.50	0.96
Amplitude of DC reference voltage	0.50 V	8.60 V
Amplitude of carrier wave	1.0 V	12.60
Modulation index of the Buck converter	0.50	0.68
Charging voltage	15.00 V	13.80 V
Efficiency	93%	85%

A comparison between the simulated and experimental results reveals notable differences in the system parameters. The simulated results indicate a supply voltage of 220 Vrms (311.12 V peak), a carrier frequency of 15 kHz, a supply frequency of 50 Hz, a buck converter duty ratio of 0.50, efficiency of 93% and a modulation index of 0.50. In contrast, the experimental results demonstrate a supply voltage of 328.68 V, a carrier frequency of 14.7 kHz, a supply frequency of 50.02 Hz, a buck converter duty ratio of 0.96, efficiency of 85% and a modulation index of 0.68. These variations can be attributed to practical non-idealities such as component tolerances, switching losses, and measurement uncertainties, which are not fully captured in the simulation environment. The comparison emphasizes the importance of experimental validation, as real-world conditions often introduce deviations from theoretical predictions, thereby providing a more accurate representation of system performance.

5 Conclusion

This system has been successfully designed and implemented by showcasing a smart battery charging system capable of addressing the major problems associated with traditional chargers such as poor monitoring, overcharging, reduced battery lifespan, and the lack of real-time user feedback. By integrating controlled charging techniques with IoT-based monitoring, the system provides a safer, more efficient and more user-friendly

charging process monitoring. The objectives of the project were achieved through the development of a microcontroller-based charger using PWM regulation, real-time sensing modules, and both local and remote display interfaces. The LCD screen allowed immediate on-device monitoring of voltage, current, and charging stages, while the Blynk application provided convenient remote supervision via the internet. Together, these features ensure accurate measurement, protection of the battery, and continuous visibility of the charging operation.

Author contributions

A.C.I: conceptualized and designed the different parts as well as simulated the work in MATLAB/Simulink software tool of the work U.H: played the role of reviewing other related work of the manuscript J.O Hyginus: handled grammatical corrections and equation modelling P.O E: was in charge of statistical analysis of the manuscript P.E: handled typographical errors' corrections in the manuscript C.U.E: carried out coordination and supervising the overall work of the manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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